Q1)

Ans) Given :- 1) $p(x|\omega_1) = N(\mu_1, \Sigma)$

2) $\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

3) $p(x|\omega_2) = N(\mu_2, \Sigma)$

4) $\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

5) $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

6) $\Lambda = \begin{bmatrix} 0 & \lambda_{12} \\ \lambda_{21} & 0 \end{bmatrix}$, $\lambda_{12} = \lambda_{21} = 1$

7) Priors are equal. $P(\omega_1) = P(\omega_2)$

R.T.F:- 1) The likelihood ratio decision rule in terms of $\omega^T x > \tau$ and find ω and τ

2) Classify $x = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix}$

Sol) 1) We 'decide $x \in \omega_1$' if $\frac{p(x|\omega_1)}{p(x|\omega_2)} > 1 \Rightarrow \frac{N(\mu_1, \Sigma)}{N(\mu_2, \Sigma)} > 1$ — ①

$$\frac{N(\mu_1, \Sigma)}{N(\mu_2, \Sigma)} = \frac{\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)}}{\frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)}}$$

$$= e^{-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)} \quad \text{— ②}$$

.5

Combining eq ① & ②, we get

$$e^{-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)} > 1$$

$$\Rightarrow \ln\left(e^{-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)}\right) > \ln 1$$

$$\Rightarrow -\frac{1}{2}\left((x-\mu_1)^T \Sigma^{-1}(x-\mu_1) - (x-\mu_2)^T \Sigma^{-1}(x-\mu_2)\right) > 0$$

$$\Rightarrow -\left((x-\mu_1)^T \Sigma^{-1}(x-\mu_1) - (x-\mu_2)^T \Sigma^{-1}(x-\mu_2)\right) > 0$$

Since $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \Rightarrow \Sigma^{-1} = I_2$ or $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow -\left((x-\mu_1)^T I_2 (x-\mu_1) - (x-\mu_2)^T I_2 (x-\mu_2)\right) > 0$$

.5

$$\Rightarrow -\left((x-\mu_1)^T (x-\mu_1) - (x-\mu_2)^T (x-\mu_2)\right) > 0$$

Since $\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (zero column vector), thus $x - \mu_1 = x$

$-(x^Tx - (x-\mu_2)^T(x-\mu_2)) > 0$

$-(x^Tx - x^Tx + x^T\mu_2 + \mu_2^Tx - \mu_2^T\mu_2) > 0$

let $x \in \mathbb{R}^2 = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ (for reference)

$-\left([a_1\ a_2]\begin{bmatrix}1\\1\end{bmatrix} + [1\ 1]\begin{bmatrix}a_1\\a_2\end{bmatrix} - [1\ 1]\begin{bmatrix}1\\1\end{bmatrix}\right) > 0$

$-2(a_1+a_2) + 2 > 0$

$a_1 + a_2 < 1$

$[a_1\ a_2]\begin{bmatrix}1\\1\end{bmatrix} < 1$ or $[1\ 1]\begin{bmatrix}a_1\\a_2\end{bmatrix} < 1$

we can multiply $-K$ on both sides (provided $K \geq 0$ otherwise sign inequality sign will flip!)

$[-K\ -K]\begin{bmatrix}a_1\\a_2\end{bmatrix} > -K$ or $w^Tx > \tau$ where $w^T = [K\ K]_{1\times 2}$ $K\tau = K$

.5 $\boxed{w = \begin{bmatrix}-K\\-K\end{bmatrix}_{2\times 1} \text{ and } \tau = -K}$

2) If $x = \begin{bmatrix}0.4\\0.2\end{bmatrix}$

we 'decide $x \in \omega_1$' if $w^Tx > \tau$

$[K\ -K]\begin{bmatrix}0.4\\0.2\end{bmatrix} > -K$

$-0.6K > -K$ (True). Hence, $x \in \omega_1$. .5

(Q2)

Ans) Given :- 1) $p(x|\omega_1) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ or $N(0,1)$

2) $p(x|\omega_2) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ where $\lambda = \frac{1}{\sqrt{2\pi}}$

3) $P(\omega_1) = P(\omega_2) = \frac{1}{2}$

4) 0-1 Loss

- R.T.F :- 1) Find decision boundary by bayes decision rule & find out regions

2) Compute $P_{Bayes}(error)$ in the form of erfc(·)

3) Derive decision boundary acc. to threshold $P(\text{decide } \omega_2 | x \in \omega_1)$ in the form of erfc(·)

4) Using decision boundary of 3 give find expression for $P(\text{decide } \omega_1 | x \in \omega_2)$..

Sol) 1) Case I :- $x < 0$

$$P(\omega_1|x) = \frac{P(x|\omega_1)\cdot P(\omega_1)}{P(x)} = \frac{\frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{2}}{P(x)} > 0 \text{ (Always)}$$

$$P(\omega_2|x) = \frac{P(x|\omega_2)\cdot P(\omega_2)}{P(x)} = \frac{0 \cdot \frac{1}{2}}{P(x)} = 0$$

Thus $P(\omega_1|x) > P(\omega_2|x) \Rightarrow$ decide $x \in \omega_1$.

Case II :- $x \geq 0$

$$P(\omega_1|x) = \frac{\left(\frac{1}{\sqrt{2\pi}} \cdot e^{-x^2/2}\right) \cdot \frac{1}{2}}{P(x)}$$

$$P(\omega_2|x) = \frac{P(x|\omega_2)\cdot P(\omega_2)}{P(x)} = \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{x}{\sqrt{2\pi}}} \cdot \frac{1}{2}}{P(x)}$$

For $P(\omega_1|x) > P(\omega_2|x)$

$$\frac{\frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{2}}{P(x)} > \frac{\frac{1}{\sqrt{2\pi}} e^{-x/\sqrt{2\pi}} \cdot \frac{1}{2}}{P(x)}$$

.5 $e^{-x^2/2} - e^{-x/\sqrt{2\pi}} > 0$ or $e^{-x^2/2} + x/\sqrt{2\pi} > 1$

$\Rightarrow \ln\left(e^{-x^2/2} + x/\sqrt{2\pi}\right) > \ln 1$

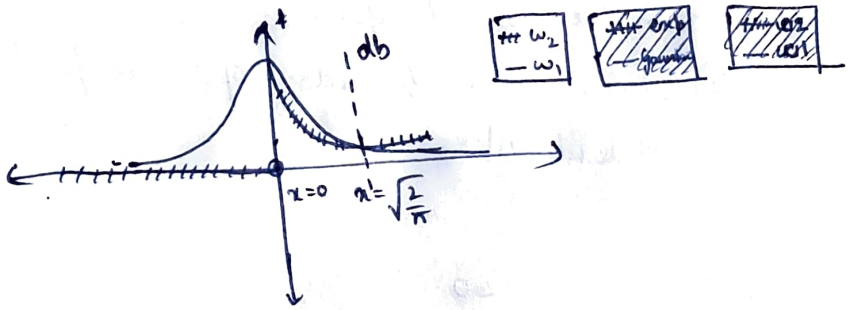
$\Rightarrow -\frac{x^2}{2} + \frac{x}{\sqrt{2\pi}} > 0$

$\Rightarrow -\frac{x}{2}\left(x - \sqrt{\frac{2}{\pi}}\right) > 0$

$\Rightarrow \frac{x}{2}\left(x - \sqrt{\frac{2}{\pi}}\right) < 0$

Thus $P(\omega_1|x) > P(\omega_2|x)$, $0 < x < \sqrt{\frac{2}{\pi}}$

1 else $P(\omega_2|x) > P(\omega_1|x)$, $\sqrt{\frac{2}{\pi}} < x < \infty$



.5 From $x \in (-\infty, \sqrt{\frac{2}{\pi}})$, decide $x \in \omega_1$

$x \in (\sqrt{\frac{2}{\pi}}, \infty)$, decide $x \in \omega_2$

2) $P(\text{error}) = \int_{-\infty}^{\infty} P(\text{error}|x)\, P(x)\, dx$

$= \int_{-\infty}^{\infty} \min\left(P(\omega_2|x), P(\omega_1|x)\right) P(x)\, dx$

.5 $= \int_{-\infty}^{\sqrt{\frac{2}{\pi}}} P(\omega_2|x)\, P(x)\, dx + \int_{\sqrt{\frac{2}{\pi}}}^{\infty} P(\omega_1|x)\, P(x)\, dx$

$= \int_{-\infty}^{\sqrt{\frac{2}{\pi}}} P(x|\omega_2)\, P(\omega_2)\, dx + \int_{\sqrt{\frac{2}{\pi}}}^{\infty} P(x|\omega_1)\, P(\omega_1)\, dx$

$= \frac{1}{2}\left(\underbrace{\int_{-\infty}^{\sqrt{\frac{2}{\pi}}} P(x|\omega_2)\, dx}_{\text{I}} + \underbrace{\int_{\sqrt{\frac{2}{\pi}}}^{\infty} P(x|\omega_1)\, dx}_{\text{II}} \right)$ — (1)

$$I = \int_{-\infty}^{\sqrt{\frac{2}{\pi}}} p(x|\omega_2)\,dx = \int_{-\infty}^{0} p(x|\omega_2)\,dx + \int_{0}^{\sqrt{\frac{2}{\pi}}} p(x|\omega_2)\,dx$$

$$= \int_{-\infty}^{0} 0\,dx + \int_{0}^{\sqrt{\frac{2}{\pi}}} \frac{1}{\sqrt{2\pi}} e^{-x/\sqrt{2\pi}}\,dx \qquad .25$$

$$= 0 + \frac{1}{\sqrt{2\pi}} \int_{0}^{\sqrt{\frac{2}{\pi}}} e^{-x/\sqrt{2\pi}}\,dx$$

$$= \frac{1}{\sqrt{2\pi}} \times \left(\frac{e^{-x/\sqrt{2\pi}}}{-1/\sqrt{2\pi}} \right)_0^{\sqrt{\frac{2}{\pi}}}$$

$$= \frac{\frac{1}{\sqrt{2\pi}} \times \left(e^{-x/\sqrt{2\pi}}\right)_0^{\sqrt{2/\pi}}}{-\frac{1}{\sqrt{2\pi}}} = \left(e^{-x/\sqrt{2\pi}}\right)_{\sqrt{\frac{2}{\pi}}}^{0}$$

$$= \left(1 - e^{-1/\pi}\right) \quad \text{—— (2)} \qquad .25$$

$$II = \int_{\sqrt{\frac{2}{\pi}}}^{\infty} p(x|\omega_1)\,dx = \int_{\sqrt{\frac{2}{\pi}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx \qquad .5$$

$$erfc(x) = \int_{t_l = x}^{t_u = \infty} \frac{2}{\sqrt{\pi}} e^{-t^2}\,dt$$

let $t = p/\sqrt{2}$

$t_l = p_l/\sqrt{2} = x$

$p_l = x\sqrt{2}$

$t_u = p_u/\sqrt{2} = \infty$ - and $dt = dp/\sqrt{2}$

$p_u = \infty$

$$= \int_{p_l = x\sqrt{2}}^{p_u = \infty} \frac{2}{\sqrt{\pi}} e^{-p^2/2} \frac{dp}{\sqrt{2}}$$

$$= \sqrt{\frac{2}{\pi}} \int_{p_l = x\sqrt{2}}^{p_u = \infty} e^{-p^2/2}\,dp = 2 \int_{x\sqrt{2}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-p^2/2}\,dp$$ (illegible)

let's put $1/\sqrt{\pi}$ in erfc(·)

$$\text{erfc}\left(\frac{1}{\sqrt{\pi}}\right) = 2\int_{\sqrt{\frac{2}{\pi}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

(Variable change after application of limits is allowed).

$$= 2\text{II}$$

Thus $\text{II} = \frac{1}{2}\text{erfc}\left(\frac{1}{\sqrt{\pi}}\right)$ — ③

.5

Hence P(error) $= \frac{1}{2}(\text{I} + \text{II})$ (Using eq ①)

$$= \frac{1}{2}\left((1-e^{-1/\pi}) + \frac{1}{2}\text{erfc}\left(\frac{1}{\sqrt{\pi}}\right)\right)$$

$$= \frac{1}{2}(1-e^{-1/\pi}) + \frac{1}{4}\text{erfc}\left(\frac{1}{\sqrt{\pi}}\right)$$

3) $P(\text{decide } \omega_2 | x \in \omega_1) < E_1$

$$P(\text{decide } \omega_2 | x \in \omega_1) = \int_t^\infty P(x|\omega_1) \cdot P(\omega_1)\, dx$$

$$= \frac{1}{2}\int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \qquad .5$$

$$= \frac{1}{2\sqrt{2\pi}} \int_t^\infty e^{-x^2/2}\, dx$$

$$= \frac{1}{4\sqrt{2}} \int_t^\infty \frac{2}{\sqrt{\pi}} e^{-x^2/2}\, dx$$

$$= \frac{1}{4\sqrt{2}} \int_{z=t/\sqrt{2}}^\infty \frac{2}{\sqrt{\pi}} e^{-z^2}\, dz\, \sqrt{2}$$

$$= \frac{1}{4} \int_{t/\sqrt{2}}^\infty \frac{2}{\sqrt{\pi}} e^{-z^2} dz$$

$$= \frac{1}{4} \operatorname{erfc}(t/\sqrt{2})$$

less than is also fine

For boundary, $\frac{1}{4}\operatorname{erfc}(t/\sqrt{2}) = E_1 \Rightarrow t = \sqrt{2}\operatorname{erfc}^{-1}(4E_1)$ .5

4) $P(\text{decide } \omega_1 | \omega_2) = \int_0^t \frac{1}{\sqrt{2\pi}} e^{-x/\sqrt{2\pi}}\, dx$ .5

$$= \frac{-1}{\sqrt{2\pi}} e^{-x/\sqrt{2\pi}} \Big|_0^t$$

$$= e^{-x/\sqrt{2\pi}} \Big|_t^0$$

$$= (1 - e^{-t/\sqrt{2\pi}}) \quad \text{where } t = \sqrt{2}\operatorname{erfc}^{-1}(4E_1) \qquad .5$$

(Q3)

Ans) R.T.F:- The conditions for Bayes and Miniman risk giving same decision boundary

Assumptions taken

1) The distribution (likelihood) is of same nature {in this case gaussian}.
2) The distribution is equivariant (have equal variance).
3) ~~The equivariant equal~~

The boundary which we get from miniman risk is the mean of distributions mean (from distribution $P(x|\omega_1) = N(\mu_1, \sigma^2)$ & $P(x|\omega_2) = N(\mu_2, \sigma^2)$

$$db_{mm} = \frac{\mu_1 + \mu_2}{2}$$

.5

The condition for getting decision boundary from bayes risk is the posteriors must be equal

$$P(\omega_1|x) = P(\omega_2|x)$$

$$P(x|\omega_1) \cdot P(\omega_1) = P(x|\omega_2) \cdot P(\omega_2)$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu_1)^2}{\sigma^2}} P(\omega_1) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu_2)^2}{\sigma^2}} P(\omega_2)$$

$$e^{-\frac{1}{2}\left(\frac{(x-\mu_1)^2}{\sigma^2} - \frac{(x-\mu_2)^2}{\sigma^2}\right)} = \frac{P(\omega_2)}{P(\omega_1)}$$

If $P(\omega_1) = P(\omega_2) = \frac{1}{2}$ (equal priors) (Condition 1)

$$\Rightarrow e^{-\frac{1}{2}\left(\frac{(x-\mu_1)^2}{\sigma^2} - \frac{(x-\mu_2)^2}{\sigma^2}\right)} = \frac{1/2}{1/2} = 1 = e^0$$

$$\Rightarrow \frac{(x-\mu_1)^2}{\sigma^2} - \frac{(x-\mu_2)^2}{\sigma^2} = 0$$

$$\Rightarrow (\mu_2 - \mu_1)(2x - \mu_1 - \mu_2) = 0$$

$$\Rightarrow \mu_2 - \mu_1 = 0 \quad \text{(May not always be True)}$$

Thus $x = \frac{\mu_1 + \mu_2}{2}$ or $db_{Bayes} = \frac{\mu_1 + \mu_2}{2}$

.5

Hence, the decision boundary from Bayes & minimax risk is same after taking two assumptions and arriving at conclusion of equal priors.

For example, let's take two distribution $N(-1, \sigma^2)$ ($P(x|\omega_1)$) and $N(1, \sigma^2)$ ($P(x|\omega_2)$) where σ^2 is a fixed, constant variance and priors for both the class is same $P(\omega_1) = P(\omega_2) = 0.5$ .5

$$db_{mm} = \frac{-1+1}{2} = 0$$

even if this specific case is given, give full marks

$$db_{Bayes} = \frac{-1+1}{2} = 0$$ (As it is given that priors are equal) .5

but this should have some of the above shown steps, if not all

$$db_{mm} = db_{bayes} = 0$$

Thus, using this example we illustrate when the assumptions and conditions are satisfied, we get same decision boundary from minimax and bayes risk.

It is 1 marks only, but answer can be given in both ways.

Q4

Given $p(x|\omega_1) \sim N(\mu_1, \sigma^2)$ $p(x|\omega_2) \sim N(\mu_2, \sigma^2)$, $\mu_2 > \mu_1$

Equal priors $\Rightarrow$ $P(\omega_1) = P(\omega_2) = 1/2$

$$P(\text{error}|x) = \min\left(P(\omega_1|x), P(\omega_2|x)\right)$$

$$P(\text{error}) = \int_{-\infty}^{+\infty} \min\left(P(\omega_1|x), P(\omega_2|x)\right) dx$$

p(x)

$$P_e = P(\omega_1)\, P(\text{decide } \omega_2|\omega_1) + P(\omega_2)\, P(\text{decide } \omega_1|\omega_2)$$

P(x|w1) P(x|w2)

All these equations are not mandatory. Do not deduct marks if they are not given

Bayes Error

*Bayes Decision rule → decide ω_1 if $P(\omega_1|x) > P(\omega_2|x)$

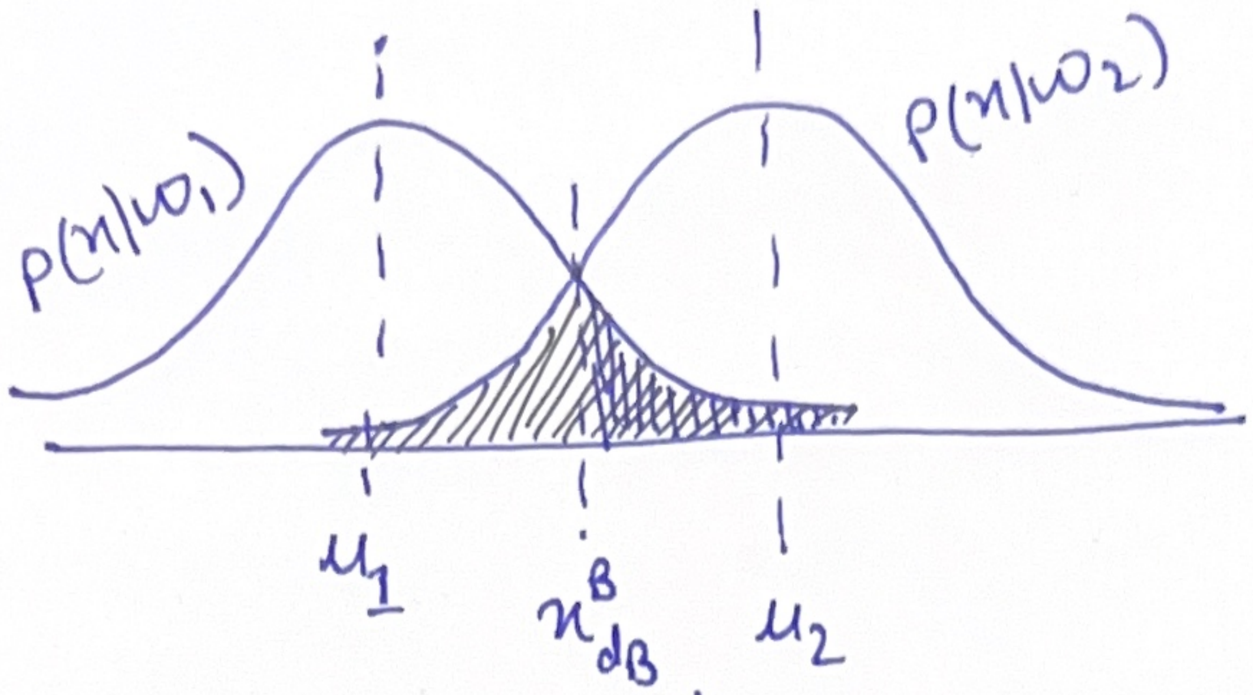
$$p(x|\omega_1)P(\omega_1) > p(x|\omega_2)P(\omega_2)$$

decision boundary → $p(x|\omega_1) = p(x|\omega_2)$ [equal priors]

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)$$

$$(x-\mu_1)^2 - (x-\mu_2)^2 = 0$$

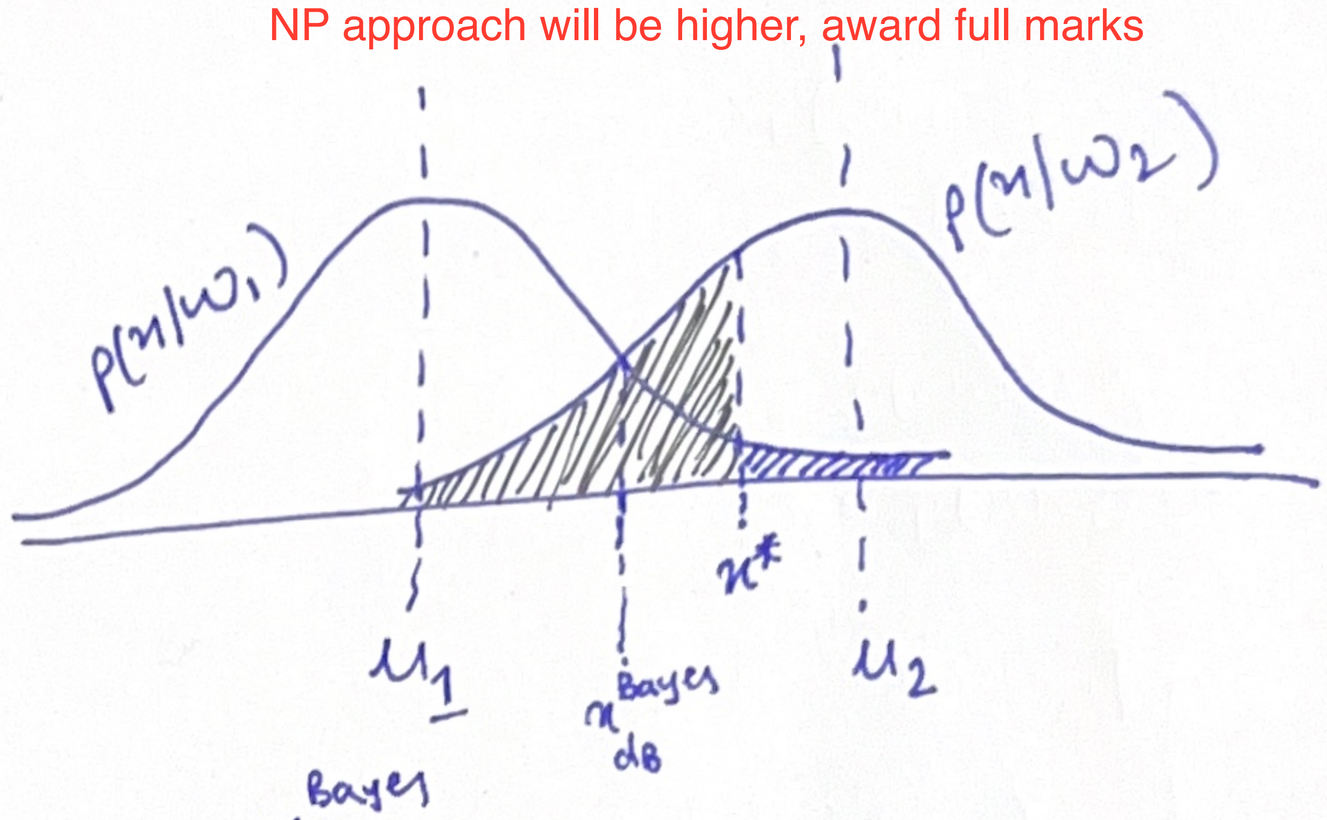
$$\hookrightarrow \boxed{x_{dB}^{\text{Bayes}} = \frac{\mu_1 + \mu_2}{2}}$$



$\left(\frac{\mu_1+\mu_2}{2}\right)$

#Bayes boundary is chosen to minimize total probability of error over all possible thresholds.

If this graph is given, with arguments that area under NP approach will be higher, award full marks



⇒ Neyman-Pearson chooses P_e^{NP} under the constraint of a budgest. $P_e^{NP} < E$

$$\Rightarrow P_e^{Bayes} = \frac{1}{2} \times \int_{x_{dB}^{Bayes}}^{\infty} p(x|\omega_1)\,dx + \frac{1}{2} \int_{-\infty}^{x_{dB}^{Bayes}} p(x|\omega_2)\,dx$$

$$\Rightarrow P_e^{NP} = \frac{1}{2} \times \int_{x^*}^{\infty} p(x|\omega_1)\,dx + \frac{1}{2} \int_{-\infty}^{x^*} p(x|\omega_2)\,dx$$

$\boxed{P_e^{Bayes} \leq P_e^{NP} \text{ always.}}$

→ Graphically we can see that the total shaded area [P(error)] increases as x^* shifts past the optimum value of $\frac{\mu_1+\mu_2}{2}$.

Note: Bayes Decision rule always gives the ~~minimum~~ minimum possible error.

From the figures as well as the expression of P_e^{Bayes} & P_e^{NP}, we can see that equality will hold when NP boundary coincides with Bayes Boundary. ($x^* = x_{dB}^{Bayes}$)

Q5

$$\eta(x) = \mathbb{P}(\omega_1|x) \in [0,1]$$

$$B_1(x) = \eta(x)^{1-\eta(x)}(1-\eta(x))^{\eta(x)}; \quad B_2(x) = \sqrt{\eta(x)(1-\eta(x))}$$

For the range $(0,1)$, $B_1(x) > 0$ and $B_2(x) > 0$.
Therefore, $B_1(x) \leq B_2(x) \iff ln(B_1(x)) \leq ln(B_2(x))$.
Let $f(x) = ln(B_2(x)) - ln(B_1(x))$.

$$ln(B_1(x)) = (1-\eta(x))\,ln(\eta(x)) + \eta(x)\,ln(1-\eta(x))$$

$$ln(B_2(x)) = \frac{1}{2}(ln(\eta(x)) + ln(1-\eta(x))) \qquad .5$$

$$f(x) = \left(\eta(x) - \frac{1}{2}\right) ln(\eta(x)) + \left(\frac{1}{2} - \eta(x)\right) ln(1-\eta(x))$$

$$f(x) = \left(\eta(x) - \frac{1}{2}\right)(ln(\eta(x)) - ln(1-\eta(x)))$$

$$f(x) = \left(\eta(x) - \frac{1}{2}\right) ln\left(\frac{\eta(x)}{1-\eta(x)}\right)$$

For $0 < \eta(x) < \frac{1}{2}$,

$$\eta(x) - \frac{1}{2} < 0, \frac{\eta(x)}{1-\eta(x)} < 1, ln\left(\frac{\eta(x)}{1-\eta(x)}\right) < 0 \implies f(x) > 0 \qquad .5$$

For $\frac{1}{2} < \eta(x) < 1$,

$$\eta(x) - \frac{1}{2} > 0, \frac{\eta(x)}{1-\eta(x)} > 1, ln\left(\frac{\eta(x)}{1-\eta(x)}\right) > 0 \implies f(x) > 0$$

even if this is not covered, do not deduct marks

For $\eta(x) = \frac{1}{2}$,

$$\eta(x) - \frac{1}{2} = 0 \implies f(x) = 0$$

For the range $(0, 1)$,

$$f(x) \geq 0$$

$$\implies ln(B_2(x)) - ln(B_1(x)) \geq 0$$

$$\implies ln(B_2(x)) \geq ln(B_1(x))$$

$$\implies B_2(x) \geq B_1(x)$$

For the range $(0, 1)$, $B_1(x)$ is a tighter bound than $B_2(x)$.
For the boundary case when $\eta(x) = 0$:

$$B_1(x) = 0^{1-0}(1-0)^0 = 0$$

$$B_2(x) = \sqrt{0(1-0)} = 0$$

For the boundary case when $\eta(x) = 1$:

$$B_1(x) = 1^{1-1}(1-1)^1 = 0$$

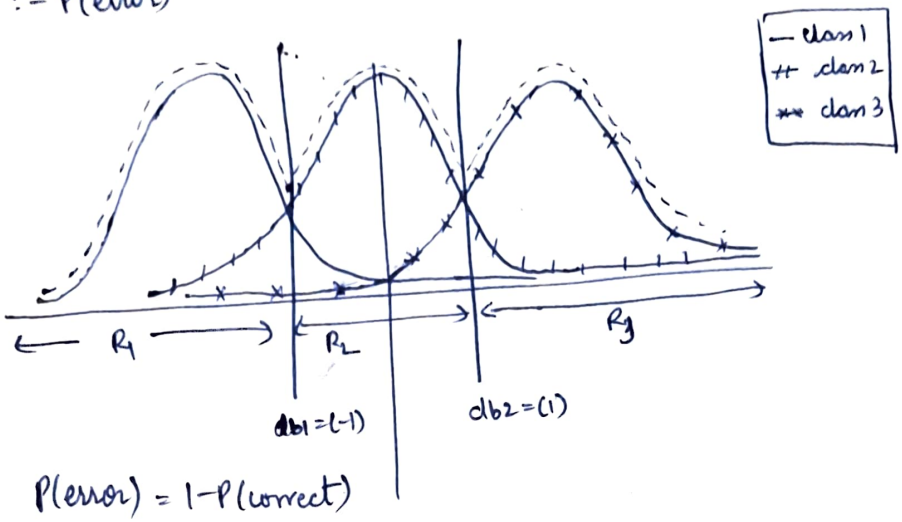
$$B_2(x) = \sqrt{1(1-1)} = 0$$

In both boundary cases, $B_1(x) \leq B_2(x)$.
Therefore, $B_1(x) \leq B_2(x)$ for the entire range, i.e. $\eta(x) \in [0, 1]$.
Thus, $B_1(x)$ is a tighter upper bound than $B_2(x)$.

Q6)

Ans) Given:- Equal Priors $P(\omega_1) = P(\omega_2) = P(\omega_3) = 1/3$

R.T.F :- P(error)



$P(error) = 1 - P(correct)$

$$P(correct) = \int_{R_1+R_2+R_3} p \max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) P(x)\,dx$$

$$= \int_{R_1} \max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) P(x)\,dx +$$

$$+ \int_{R_2} \max\left(P(\omega_2|x), P(\omega_2|x), P(\omega_3|x)\right) P(x)\,dx$$

$$+ \int_{R_3} \max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) P(x)\,dx$$

In R_1, $\max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) = P(\omega_1|x)$

In R_2, $\max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) = P(\omega_2|x)$

In R_3, $\max\left(P(\omega_1|x), P(\omega_2|x), P(\omega_3|x)\right) = P(\omega_3|x)$

$$P(correct) = \int_{R_1} P(\omega_1|x)\, P(x)\,dx + \int_{R_2} P(\omega_2|x)\, P(x)\,dx + \int_{R_3} P(\omega_3|x)\, P(x)\,dx$$

.5

$$= P(\omega)\left(\int_{R_1} P(x|\omega_1)\,dx + \int_{R_2} P(x|\omega_2)\,dx + \int_{R_3} P(x|\omega_3)\,dx\right)$$

$$= P(\omega)\left(\int_{-\infty}^{-1} P(x|\omega_1)\,dx + \int_{-1}^{1} P(x|\omega_2)\,dx + \int_{1}^{\infty} P(x|\omega_3)\,dx\right)$$

$$= 1/3\left(\int_{-\infty}^{-1} P(x|\omega_1)\,dx + \int_{-1}^{1} P(x|\omega_2)\,dx + \int_{1}^{\infty} P(x|\omega_3)\,dx\right)$$

.5

$$P(error) = 1 - P(correct) = 1 - 1/3\left(\int_{-\infty}^{-1} P(x|\omega_1)\,dx + \int_{-1}^{1} P(x|\omega_2)\,dx + \int_{1}^{\infty} P(x|\omega_3)\,dx\right)$$

Another method to find P(error) is

$P(\omega_1|x)$ is maximum from $(-\infty, -1)$, thus for region $(-1, \infty)$ it will considered as error probability

~~P(error from class 1)~~

$$P(\text{error } 1) = \int_{-1}^{\infty} P(\omega_1|x)\, p(x)\, dx$$

Illly $P(\omega_2|x)$ is maximum from $(-1, 1)$, hence it will be considered as error probability from $(-\infty, -1)$ & $(1, \infty)$

$$P(\text{error } 2) = \int_{-\infty}^{-1} P(\omega_2|x)\, p(x)\, dx + \int_{1}^{\infty} P(\omega_2|x)\, p(x)\, dx$$

Illly $P(\omega_3|x)$ is maximum from $(1, \infty)$, hence from region $(-\infty, 1)$ $P(\omega_3|x)$ will be considered in error probability

$$P(\text{error } 3) = \int_{-\infty}^{1} P(\omega_3|x) p(x)\, dx$$

Thus $P(\text{error}) = P(\text{error } 1) + P(\text{error } 2) + P(\text{error } 3)$

$$= \int_{-1}^{\infty} P(\omega_1|x)\, p(x)\, dx + \int_{-\infty}^{-1} P(\omega_2|x)\, p(x)\, dx + \int_{1}^{\infty} P(\omega_2|x)\, p(x)\, dx + \int_{-\infty}^{1} P(\omega_3|x)\, p(x)\, dx$$

$$= P(\omega) \left(\int_{-1}^{\infty} P(x|\omega_1)\, dx + \int_{-\infty}^{-1} P(x|\omega_2)\, dx + \int_{1}^{\infty} P(x|\omega_2)\, dx + \int_{-\infty}^{1} P(x|\omega_3)\, dx \right)$$

$$= \frac{1}{3} \left(\int_{-1}^{\infty} P(x|\omega_1)\, dx + \int_{-\infty}^{-1} P(x|\omega_2)\, dx + \int_{1}^{\infty} P(x|\omega_2)\, dx + \int_{-\infty}^{1} P(x|\omega_3)\, dx \right)$$